HARVARD MEDICAL SCHOOL | Master of Science in Clinical Research

SMCR Capstone Progress Report 1: Research Question and Project Scope

***Meeting deadline**:* Friday, August 28, 2026

Next milestone: Research Question Submission: Monday, September 7, 2026

Purpose: To confirm that the proposed capstone research question is clinically meaningful, sufficiently focused, methodologically answerable, and feasible within the capstone timeline.

Instructions: The student should complete Sections A-C and submit this report to the advisor at least 24 hours before the meeting. The advisor should complete Section D following the meeting.

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A. Student and Project Information

Student: Jessica Wang

Program Advisor: Dr. Jamie Robertson

Date of Meeting: August 27, 2026

Working Capstone Title:
Mechanical and optical properties of directly 3D-printed versus thermoformed orthodontic clear aligners after aqueous and beverage immersion aging: a systematic review and meta-analysis of in-vitro studies

Proposed Research Question(s):
In orthodontic clear aligners, how do directly 3D-printed compared with thermoformed appliances differ in mechanical and optical properties after immersion in water, artificial saliva, or staining beverages under clinically relevant simulation conditions?

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B. Research Question Framework

Complete the framework most appropriate for your research question.

Population: Clear orthodontic aligner materials

Intervention or exposure: Direct 3D-printed aligner materials

Comparator: Thermoformed aligner materials

Primary outcome(s): Effects of simulated oral conditions on mechanical properties

Secondary outcome(s), if applicable: Effects of simulated oral conditions on surface and optical stability

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C. Student Pre-Meeting Progress Report

1. Clinical significance and rationale

Briefly explain why this research question is important to patients, clinicians, health systems, or clinical practice.

Directly 3D-printed aligners have emerged as an alternative to conventional thermoformed aligners, offering design flexibility, in-office fabrication, and novel material behaviors. However, their clinical performance depends on properties that evolve within the intraoral environment.

2. Preliminary evidence gap

What gap, uncertainty, inconsistency, or unanswered question in the existing literature will this review address?

Studies comparing the mechanical properties of the two groups of aligner materials under water or artifical saliva immersion have grown in the past year. In terms of esthetics, which is an important aspect of clear aligners, studies on color stability were limited and have been updated in the past year.

3. Preliminary literature search

Databases searched:

PubMed, Embase, Web of Science, EBSCOhost

Search terms or concepts used:

"aligner" AND "thermoformed" AND "direct 3D printed"

Approximate number of potentially relevant studies identified:

About 20

Were closely related systematic reviews or protocols identified? [ ] No [v] Yes

If yes, explain how the proposed capstone will differ:

Since these reviews, a distinct body of work has examined mechanical properties and color stability under simulated oral conditions, which offer an opportunity for a new review in this emerging topic.

4. Current barriers or concerns

Identify any scientific, methodological, logistical, or time-management concerns.

Limited poolable data for meta-analysis

5. Guidance needed from the advisor

What specific questions or decisions would you like to address during this meeting?

Is the differentiation of my review distinct enough from the recent review published earlier this year in the *European Journal of Orthodontics?*

5. Guidance needed from the advisor

What specific questions or decisions would you like to address during this meeting?

Section D. Advisor Assessment: Overall Progress Determination

Green – On track: The project is appropriately scoped, and the student is prepared for the next milestone.

Yellow – Support required: Correctable deficiencies remain. Specific actions and a follow-up date are required.

Red – At risk: The question is not yet feasible or sufficiently developed. A formal remediation plan and program notification are required.

Overall advisor comments:

Advisor's Signature **Date**